

A

Progress Report

On

“Name of Project Title”

Submitted by

Name (Roll No.)

Submitted to

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Abstract

The objective of the application FitElevate, which contains various health related tools, can be achieved by combining features that are as follows:

1. **Using the wearable devices to track body count.** This can be achieved using various sensors like to monitor heart rate we are accessing optical heart rate sensor, to track pulse rate and sleep we are using bioimpedance sensor. We will access this data from the wearable of the user.
2. **To use various type of attributes for the formulation of BMI.** It is a reliable indicator of body fatness for most people. It is used to screen for weight categories that may lead to health problems. The BMI will be calculated using this formula:

$$\text{BMI} = \text{weight (kg)} / [\text{height (m)}]^2$$

3. **To Develop an android app to promote individual's fitness.** Developing an Android app for a fitness app can be an effective way to reach a wide audience of potential users and provide them with a convenient and accessible tool for promoting healthy behaviours.

Introduction

FitElevate is an innovative Android fitness application designed to help users reach their fitness goals and live a healthier life. With its intuitive interface, FitElevate provides a comprehensive range of features that cater to the fitness needs of users of all levels.

Our application will contain features like heart rate monitor, steps counter, BMI, BMR, exercises for beginners, intermediate and advanced level of individuals and yoga. By combining all these features, we will develop our application bundled with these advanced and essential tools to live a happier & healthier life.

Overall, FitElevate is a comprehensive and user-friendly fitness application that provides everything users need to reach their fitness goals. With its personalized workout generator, food tracking system, and progress tracking system, FitElevate is the ultimate tool for anyone looking to improve their fitness and live a healthier life.

Literature Survey ^[1]

Here three fitness related applications are reviewed: *Nike Training Club*, *Fit Body*, and *Le Sweat TV*. For each of them, an account is registered to test the major features to point out the strengths and weaknesses.

Nike Training Club: Work out with Nike trainers live or on-demand with the Nike Training Club app. After downloading the app, you'll get customized workout suggestions based on the results of a quick quiz. The app offers a range of exercise styles, including high intensity interval training (HIIT), yoga, core, cardio, and strength training. You can also search by muscle group, workout focus, and equipment to find a class.

Pros

- free to use.
- wide selection of workouts.
- multiple filters, including muscle group, equipment available, and duration
- warmups and cooldowns included.

Cons

- takes up a decent amount of storage space on your phone.
- can't play your own music while streaming classes.
- quick transitions and fast-paced moves may be difficult for beginners.

Fit Body: A community created for women by women. Fit Body features 12 programs led by four expert trainers to help you reach your fitness goals. The classes are designed to be 30 minutes or shorter. The app offers 72 weeks of meal plans with custom portion recommendations according to your daily calorie needs, plus access to more than 300 recipes to create your own meal plan. There's also a community function within the app, allowing you to connect with others on your wellness journey.

Pros

- offers nutrition tips and meal plans.
- tracks workouts and mental/emotional wellness
- includes alternative move suggestions during workouts.

Cons

- may not be challenging enough for some.

Le Sweat TV: Created by Charlee Atkins, CPT, Le Sweat TV essentially gives you a personal trainer at your fingertips. During each workout, Atkins is right there with you, providing helpful cues and modifications for all fitness levels. You can browse the 250-plus workouts by body part, equipment needed, or desired class length (5–30 minutes) to get started. You can also stream the videos on demand or download them for sweat sessions on the go.

Pros

- trainer-led classes
- classes don't require a lot of equipment.

Cons

- workout library not as robust as others on this list
- limited workout styles
- lacks longer classes.

Comparison with apps above mentioned: In the comparison with other fitness apps, FitElevate has advantages on these features which are as follows:

- Exercises can be categorized in our app to make it more user-oriented product so that there should not be any difficulty for the beginners.
- There are bunch of exercises provided in our application so there is no limitation in workout styles.
- There is support for robust workout libraries.
- Exercises are user oriented so that makes it equally challenging at every step.

Proposed Work

XYZ.....

System Requirement Specification

Hardware Requirements [\[2\]](#)

Client

- Smartphone with processor 4.4 Snapdragon 450 or higher.
- RAM : 4 GB
- Storage : 8 GB

Developer

- Processor : Core i5-8400 3.0 GHz
- RAM : 8 GB
- Storage : 100 GB

Software Requirements

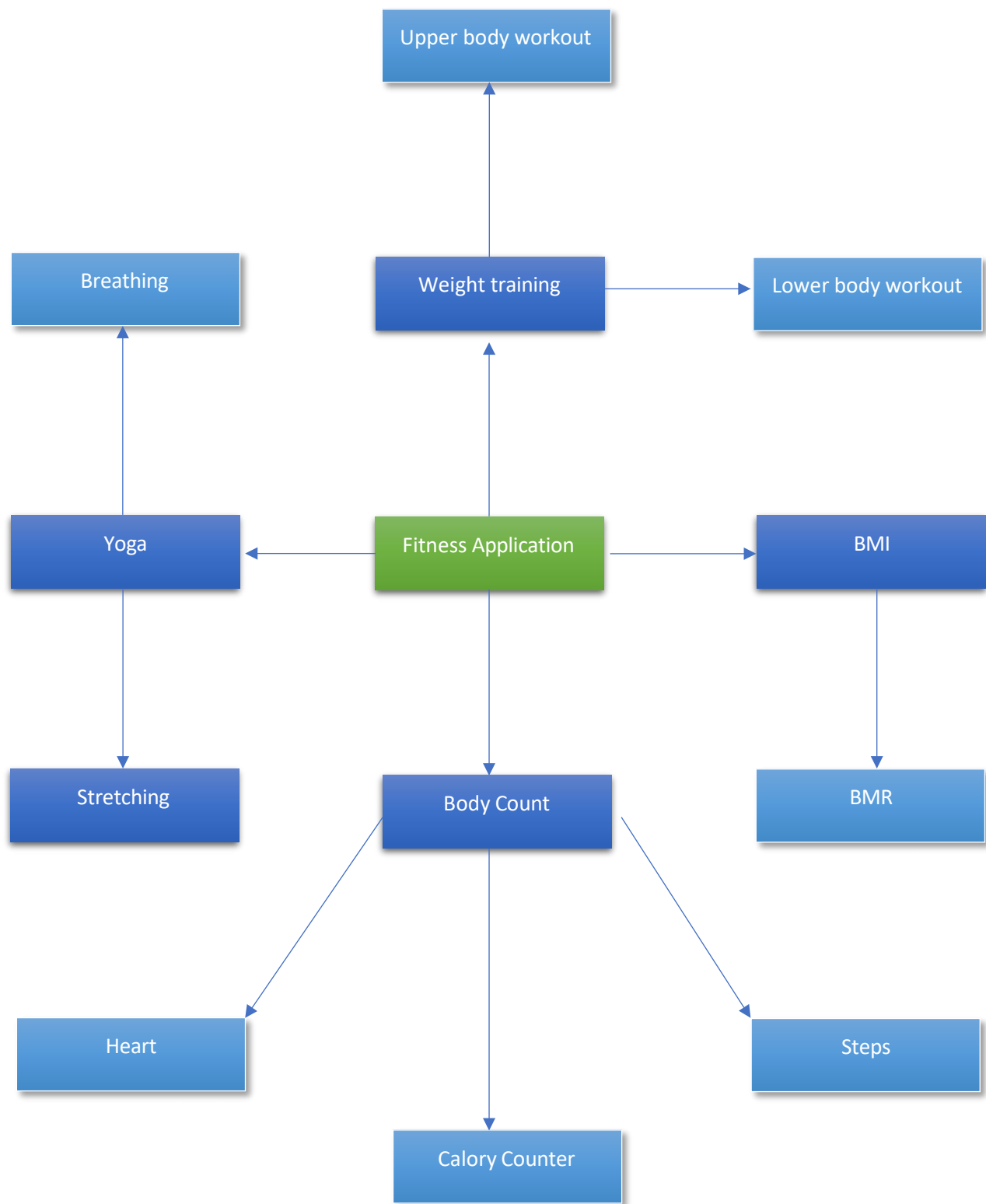
Client

- Operating System : Android 7 or above

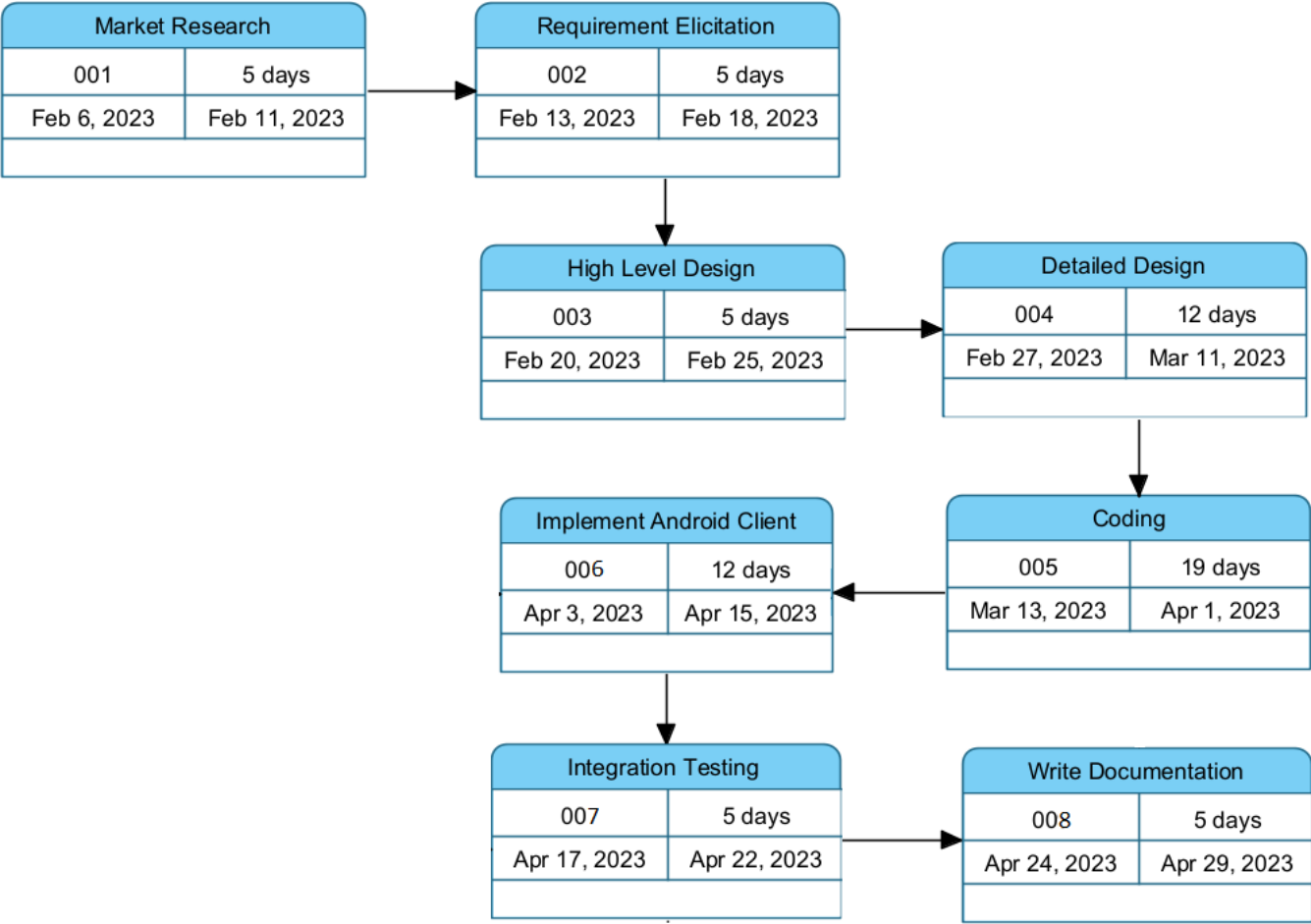
Developer

- Development Environment : Android Studio
- Database : SQLite
- Operating System : Windows 10
- Documentation tool : MS Word, MS Powerpoint

Proposed Tentative Architecture



PERT Chart ^[3]



Conclusion and Future Scope

References

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